Printix MCP Server — Tool Reference

Complete catalogue of MCP tools exposed by the Printix MCP Server for AI assistants.

> **Version:** 6.7.107 | **Transport:** Streamable HTTP (`/mcp` for claude.ai) + SSE (`/sse` for ChatGPT) | **Auth:** OAuth 2.0 Bearer Token (per tenant) | **Public Endpoint:** `https://mcp.printix.cloud`

1. Status & Info

Tool	Description
`printix_status()`	Returns server version, configured tenant, OAuth app health. The sanity-check tool — always call this first when debugging.

2. Printers & Queues

Tool	Description
`printix_list_printers(search, page, size)`	List printers in the tenant. Supports free-text search (name, model, location).
`printix_get_printer(printer_id, queue_id)`	Full detail of a single printer/queue pair, including capabilities.

3. Print Jobs

Tool	Description
`printix_list_jobs(queue_id, page, size)`	Current print jobs in the tenant. Optional filter by queue.
`printix_get_job(job_id)`	Detail of one print job (status, owner, size, creation time).
`printix_submit_job(...)`	Submit a new print job. Returns upload URL for the 3-step flow (submit → upload → complete).
`printix_complete_upload(job_id)`	Finalize a print job after the file has been uploaded to the pre-signed URL.
`printix_delete_job(job_id)`	Cancel or delete a print job.
`printix_change_job_owner(job_id, new_owner_email)`	Transfer Secure-Print ownership to another user (e.g. for delegation).

4. Card Management (Printix Cloud)

Tool	Description
`printix_list_cards(user_id)`	All cards of one user, **enriched** with local DB mappings (raw UID, transform profile, notes, decoded hex/decimal variants).
`printix_search_card(card_id, card_number)`	Lookup a single card by Printix ID or physical card number.

`printix_register_card(user_id, card_number)`	Register a card directly in Printix (skips the local mapping layer).
`printix_delete_card(card_id)`	Remove a card from Printix.

5. Card Management (Local DB)

Tool	Description
`printix_list_card_profiles()`	All transform profiles — built-in vendor presets (HP, Konica, Ricoh etc.) plus tenant-custom profiles.
`printix_get_card_profile(profile_id)`	Single profile with its rules JSON (how a raw UID gets transformed into the Printix secret).
`printix_search_card_mappings(search, printix_user_id)`	Full-text search across locally stored mappings.
`printix_get_card_details(card_id, card_number)`	Enriched single-card view combining Printix API + local DB data.
`printix_decode_card_value(card_value)`	Decode a raw Printix base64 secret into hex / decimal / reversed variants — useful for debugging.
`printix_transform_card_value(card_value, ...)`	Dry-run: apply a transform profile to a raw UID without registering.
`printix_get_user_card_context(user_id)`	One-shot combo: user details + all cards (enriched) + all local mappings for that user.

6. Users

Tool	Description
`printix_list_users(role, query, page, page_size)`	List users. Default `role='USER'`. Call again with `'GUEST_USER'` for guest accounts.
`printix_get_user(user_id)`	Full user detail including attributes and group memberships.
`printix_create_user(email, display_name, pin, password)`	Create a guest user account. PIN must be exactly 4 digits.
`printix_delete_user(user_id)`	Delete a guest user. Managers can't be deleted via API.
`printix_generate_id_code(user_id)`	Issue a fresh 6-digit ID code for card-reader fallback authentication.

7. Groups

Tool	Description
`printix_list_groups(search, page, size)`	List user groups (used for printer assignment).
`printix_get_group(group_id)`	Single group with members and permissions.
`printix_create_group(name, external_id)`	Create a group. `external_id` typically matches your directory source.
`printix_delete_group(group_id)`	Delete a group.

8. Workstations

Tool	Description
`printix_list_workstations(search, page, page_size)`	List endpoints with Printix Client installed. Filter by hostname or user.
`printix_get_workstation(workstation_id)`	Workstation detail — OS, last seen, online state, associated user.

9. Sites & Networks

Tool	Description
`printix_list_sites(search, page, size)`	List sites (logical locations like "HQ Copenhagen").
`printix_get_site(site_id)`	Site detail with networks and printers.
`printix_create_site(name, path, ...)`	Create a new site node in the organizational tree.
`printix_update_site(site_id, ...)`	Modify site metadata.
`printix_delete_site(site_id)`	Delete a site (requires empty networks).
`printix_list_networks(site_id, page, size)`	List networks (subnets where printers live). Optionally scoped to a site.
`printix_get_network(network_id)`	Network detail including CIDR, SNMP config, printers.
`printix_create_network(name, ...)`	Create a network (subnet + SNMP binding).
`printix_update_network(network_id, ...)`	Modify network.
`printix_delete_network(network_id)`	Delete a network.

10. SNMP Configurations

Tool	Description
`printix_list_snmp_configs(page, size)`	All SNMP configurations (v2c/v3 credentials for printer discovery).
`printix_get_snmp_config(config_id)`	Single SNMP config.
`printix_create_snmp_config(name, ...)`	Create an SNMP config (v2c with community or v3 with user/auth/priv).
`printix_delete_snmp_config(config_id)`	Delete SNMP config.

11. Reporting — Queries

Tool	Description
`printix_reporting_status()`	Health of the SQL Server data source, last ingest time, row counts.
`printix_query_print_stats(...)`	Raw print-volume aggregation. Group by user / printer / site / day.

`printix_query_cost_report(...)`	Cost calculation with configurable €/page for mono, color, sheet.
`printix_query_top_users(start_date, end_date, top_n, metric, ...)`	Ranking of most active users. Metrics: pages, cost, jobs, color_pages.
`printix_query_top_printers(start_date, end_date, top_n, metric, ...)`	Ranking of most-used printers — same metrics as top_users.
`printix_query_anomalies(...)`	Detect usage spikes or drops vs. historical baseline.
`printix_query_trend(...)`	Time-series for print volume, grouped by day / week / month.
`printix_query_any(sql_like_spec)`	Generic escape hatch — execute a saved-template spec directly.

12. Reporting — Templates & Schedules

Tool	Description
`printix_save_report_template(...)`	Persist a configured report for re-use + scheduling.
`printix_list_report_templates()`	All saved report templates in the tenant.
`printix_get_report_template(report_id)`	Template detail with its query spec and design options.
`printix_delete_report_template(report_id)`	Delete template.
`printix_run_report_now(report_id, report_name)`	Execute a template ad-hoc and return the resulting data.
`printix_send_test_email(recipient)`	Verify SMTP configuration / email delivery.
`printix_schedule_report(...)`	Attach a cron-like schedule to a template (daily/weekly/monthly).
`printix_list_schedules()`	All active report schedules.
`printix_delete_schedule(report_id)`	Remove schedule from a template.
`printix_update_schedule(...)`	Modify an existing schedule.

13. Reporting — Design & Preview

Tool	Description
`printix_list_design_options()`	Available chart types and layout presets for PDF reports.
`printix_preview_report(...)`	Render a report preview (HTML/PDF) without sending it by email.

14. Demo Data

Tool	Description
`printix_demo_setup_schema()`	Create demo tables in SQLite if missing. Safe to call repeatedly.

`printix_demo_generate(...)`	Generate synthetic print-job data — useful for showcasing dashboards without real data.
`printix_demo_rollback(demo_tag)`	Remove a demo batch by tag.
`printix_demo_status()`	List active demo datasets and row counts.

15. Audit Log & Feature Requests

Tool	Description
`printix_query_audit_log(start_date, end_date, action_prefix, object_type, limit)`	Query the tenant audit log — user actions, credential changes, approvals, feature requests.
`printix_list_feature_requests(status, limit)`	List user-submitted feature requests / ideas.
`printix_get_feature_request(ticket_id)`	Detail of a specific feature request.

16. Backup Management

Tool	Description
`printix_list_backups()`	Existing DB + secrets backups stored under `/config/backups`.
`printix_create_backup()`	Create a fresh backup (manual on-demand).

17. Capture Profiles

Tool	Description
`printix_list_capture_profiles()`	Capture webhook profiles (OCR and forwarding pipelines for scanned documents).
`printix_capture_status()`	Capture subsystem health plus recent webhook activity.

18. Site & Network Aggregations

Tool	Description
`printix_site_summary(site_id)`	Site overview bundling networks, printers and SNMP in one response.
`printix_network_printers(network_id, site_id)`	Printers on a specific network.
`printix_get_queue_context(queue_id, printer_id)`	Full context for a queue (printer + site + network + SNMP).
`printix_get_network_context(network_id)`	Network + site + printers + SNMP in one call.
`printix_get_snmp_context(config_id)`	SNMP config plus networks/sites/printers that use it.

19. Cross-Source Insights NEW v6.7.107

Tool	Description
`printix_find_user(query)`	Fuzzy-match a user via email, name, or ID across USER + GUEST_USER roles. Pre-step for anything that needs a `user_id`.
`printix_user_360(query)`	Complete profile: user details + all cards (enriched) + associated workstations. Single-call answer to "what's the state of user X?".
`printix_printer_health_report()`	Online/offline breakdown grouped by site, including the list of offline printers for quick triage.
`printix_tenant_summary()`	Executive dashboard — counts for users, printers, workstations, groups, local cards. Perfect for "give me an org overview".
`printix_diagnose_user(email)`	Heuristic troubleshooting — finds missing cards, wrong role types, orphaned accounts. Returns findings with severities + suggestions.

20. Card Management — Tenant-Wide NEW v6.7.107

Tool	Description
`printix_list_cards_by_tenant(status)`	All cards across all users. Filter: `mapped` / `unmapped` / `all`.
`printix_find_orphaned_mappings()`	Local mappings that point to Printix cards which no longer exist (leftovers from out-of-app deletes).
`printix_bulk_import_cards(csv_data, profile_id, dry_run)`	CSV mass import (`email,card_uid[,notes]`) with transform-profile application and dry-run mode.
`printix_suggest_profile(sample_uid)`	Tries all profiles against a sample UID and ranks the best match.
`printix_card_audit(user_email)`	Per-user card audit trail with source, notes, timestamps.

21. Print Jobs & Reporting — High-Level NEW v6.7.107

Tool	Description
`printix_top_printers(days, limit, metric)`	Most-used printers in last N days. Convenience wrapper with auto-computed date range.
`printix_top_users(days, limit, metric)`	Most-active users in last N days. Wrapper around query_top_users.
`printix_jobs_stuck(minutes)`	Jobs older than N minutes still stuck in queue — input for alerting/monitoring.
`printix_print_trends(group_by, days)`	Print-volume time series grouped by day/week/month.

`printix_cost_by_department(department_field, days, cost_per_mono, cost_per_color)`	Cost aggregation by user-attribute field. Requires Printix to expose the department attribute.
`printix_compare_periods(days_a, days_b, offset_b)`	Compares two equal-length windows — e.g. "last 30 days vs. the 30 before that".

22. Access & Governance NEW v6.7.107

Tool	Description
`printix_list_admins()`	All SYSTEM_MANAGER / SITE_MANAGER / KIOSK_MANAGER accounts. Some manager roles are API-hidden — empty results may still be valid.
`printix_permission_matrix()`	User × Group matrix (groups drive printer access in Printix).
`printix_inactive_users(days)`	Users with no activity in last N days — cleanup candidates. Depends on `lastSignIn` exposure.
`printix_sso_status(email)`	Entra/Azure SSO link status for a user, best-effort based on available API fields.

23. Agent Workflow Helpers NEW v6.7.107

Tool	Description
`printix_explain_error(code_or_message)`	Plain-language explanation + fix suggestions for known error codes (no_printix_user, 409, etc.) from internal knowledge base.
`printix_suggest_next_action(context)`	Heuristic advisor — given a situation, suggests which tools to call next.
`printix_send_to_user(user_email, file_url OR file_content_b64, filename, target_printer, copies)`	High-level composite: resolve printer → submit → upload → complete → change owner. Accepts URL or base64 content.

`printix_onboard_user(email, display_name, role, pin, password, groups)`	Full onboarding workflow in one call. Currently creates guest accounts; native SSO users go through Entra.
`printix_offboard_user(email, force)`	Leaver flow: delete cards (Printix + local mappings), then delete account. Force flag required for non-guest users.

24. Quality of Life NEW v6.7.107

Tool	Description
`printix_whoami()`	Current tenant context plus configured OAuth scopes. Debug helper for "am I talking to the right tenant?".
`printix_quick_print(recipient_email, file_url, filename)`	One-shot send: first available printer + default settings. For "just print this for Marcus" flows.
`printix_resolve_printer(name_or_location)`	Fuzzy-match a printer by name/location/model → `printer_id:queue_id`.
`printix_natural_query(question)`	Maps a natural-language question to relevant reporting tools (hint tool, not an LLM).

Conventions

- **Return format:** every tool returns a JSON-encoded string.
- **Error shape:** Printix API errors return `{"error": true, "status_code": N, "message": "...", "error_id": "..."}`. Other errors return `{"error": "message"}`.
- **Tenant isolation:** each request resolves its tenant via Bearer token. Tools operate only on that tenant's data.
- **Pagination:** listing tools accept `page` (0-based) and `page_size` / `size` (default 50, max 200).
- **Date format:** `YYYY-MM-DD` for all date arguments.
- **Cost parameters:** `cost_per_mono`, `cost_per_color`, `cost_per_sheet` are per page/sheet in the tenant's currency; defaults reflect rough EU averages.

Integration Patterns

"What cards does Marcus have?"

```
printix_find_user("marcus")
    → user_id
```

```
printix_get_user_card_context(user_id)
    → cards + enriched local data
```

"Who printed the most last month?"

```
printix_top_users(days=30, limit=10, metric="pages")
```

"User X can't print — help me debug"

```
printix_diagnose_user("x@company.com")
    → findings list with severities + suggestions
```

"Onboard a new employee"

```
printix_onboard_user(email=..., display_name=..., groups="grp1,grp2")
printix_generate_id_code(user_id)   # optional
# card registration happens in iOS/Web client
```

"How did print volume change vs last month?"

```
printix_compare_periods(days_a=30, days_b=30)
```

"Just print this file for Marcus"

```
printix_quick_print(
    recipient_email="marcus@company.com",
    file_url="https://.../document.pdf",
    filename="document.pdf")
```